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SIGN LANGUAGE INTERPRETATION WITH COLLABORATIVE AGENTS

Abstract

A method for recognizing sign language using collaborative augmented reality devices is described. In one aspect, a method includes accessing a first image generated by a first augmented reality device and a second image generated by a second augmented reality device, the first image and the second image depicting a hand gesture of a user of the first augmented reality device, synchronizing the first augmented reality device with the second augmented reality device, in response to the synchronizing, distributing one or more processes of a sign language recognition system between the first and second augmented reality devices, collecting results from the one or more processes from the first and second augmented reality devices, and displaying, in near real-time in a first display of the first augmented reality device, text indicating a sign language translation of the hand gesture based on the results.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation of U.S. patent application Ser. No. 17/967,209, filed Oct. 17, 2022, which is incorporated by reference herein in its entirety.

TECHNICAL FIELD

[0002] The subject matter disclosed herein generally relates to an augmented reality (AR) device. Specifically, the present disclosure addresses systems and methods for sign language interpretation using collaborative agents.

BACKGROUND

[0003] An augmented reality (AR) device enables a user to observe a scene while simultaneously seeing relevant virtual content that may be aligned to items, images, objects, or environments in the field of view of the device. A virtual reality (VR) device provides a more immersive experience than an AR device. The VR device blocks out the field of view of the user with virtual content that is displayed based on a position and orientation of the VR device.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

[0004] To easily identify the discussion of any particular element or act, the most significant digit or digits in a reference number refer to the figure number in which that element is first introduced.

[0005] FIG. 1 is a block diagram illustrating a network environment for sign language translation using collaborating augmented reality devices in accordance with one example embodiment.

[0006] FIG. 2 is a block diagram illustrating a network environment for sign language translation using collaborating augmented reality devices in accordance with one example embodiment.

[0007] FIG. 3 is a block diagram illustrating an augmented reality device in accordance with one example embodiment.

[0008] FIG. 4 is a block diagram illustrating a collaborative sign language system in accordance with one example embodiment.

[0009] FIG. 5 is an interaction diagram illustrating interactions between collaborative augmented reality devices in accordance with one example embodiment.

[0010] FIG. 6 is an interaction diagram illustrating interactions between collaborative augmented reality devices in accordance with one example embodiment.

[0011] FIG. 7 is an interaction diagram illustrating interactions between collaborative augmented reality devices in accordance with one example embodiment.

[0012] FIG. 8 is an interaction diagram illustrating interactions between collaborative augmented reality devices in accordance with one example embodiment.

[0013] FIG. 9 is a flow diagram illustrating a method for sign language translation in accordance with one example embodiment.

[0014] FIG. 10 is a flow diagram illustrating a method for distributing sign language translation processes in accordance with one example embodiment.

[0015] FIG. 11 illustrates a network environment in which a head-wearable device can be implemented according to one example embodiment.

[0016] FIG. 12 is block diagram showing a software architecture within which the present disclosure may be implemented, according to an example embodiment.

[0017] FIG. 13 is a diagrammatic representation of a machine in the form of a computer system within which a set of instructions may be executed for causing the machine to perform any one or more of the methodologies discussed herein, according to one example embodiment.

DETAILED DESCRIPTION

[0018] The description that follows describes systems, methods, techniques, instruction sequences, and computing machine program products that illustrate example embodiments of the present subject matter. In the following description, for purposes of explanation, numerous specific details are set forth in order to provide an understanding of various embodiments of the present subject matter. It will be evident, however, to those skilled in the art, that embodiments of the present subject matter may be practiced without some or other of these specific details. Examples merely typify possible variations. Unless explicitly stated otherwise, structures (e.g., structural components, such as modules) are optional and may be combined or subdivided, and operations (e.g., in a procedure, algorithm, or other function) may vary in sequence or be combined or subdivided.

[0019] The term “augmented reality” (AR) is used herein to refer to an interactive experience of a real-world environment where physical objects that reside in the real-world are “augmented” or enhanced by computer-generated digital content (also referred to as virtual content or synthetic content). AR can also refer to a system that enables a combination of real and virtual worlds, real-time interaction, and 3D registration of virtual and real objects. A user of an AR system perceives virtual content that appears to be attached or interact with a real-world physical object.

[0020] The term “virtual reality” (VR) is used herein to refer to a simulation experience of a virtual world environment that is completely distinct from the real-world environment. Computer-generated digital content is displayed in the virtual world environment. VR also refers to a system that enables a user of a VR system to be completely immersed in the virtual world environment and to interact with virtual objects presented in the virtual world environment.

[0021] The term “AR application” is used herein to refer to a computer-operated application that enables an AR experience. The term “VR application” is used herein to refer to a computer-operated application that enables a VR experience. The term “AR/VR application” refers to a computer-operated application that enables a combination of an AR experience or a VR experience.

[0022] The term “visual tracking system” is used herein to refer to a computer-operated application or system that enables a system to track visual features identified in images captured by one or more cameras of the visual tracking system. The visual tracking system builds a model of a real-world environment based on the tracked visual features. Non-limiting examples of the visual tracking system include: a visual Simultaneous Localization and Mapping system (VSLAM), and Visual Inertial Odometry (VIO) system. VSLAM can be used to build a target from an environment, or a scene based on one or more cameras of the visual tracking system. A VIO system (also referred to as a visual-inertial tracking system) determines a latest pose (e.g., position and orientation) of a device based on data acquired from multiple sensors (e.g., optical sensors, inertial sensors) of the device.

[0023] The term “Inertial Measurement Unit” (IMU) is used herein to refer to a device that can report on the inertial status of a moving body including the acceleration, velocity, orientation, and position of the moving body. An IMU enables tracking of movement of a body by integrating the acceleration and the angular velocity measured by the IMU. IMU can also refer to a combination of accelerometers and gyroscopes that can determine and quantify linear acceleration and angular

velocity, respectively. The values obtained from the IMUs gyroscopes can be processed to obtain the pitch, roll, and heading of the IMU and, therefore, of the body with which the IMU is associated. Signals from the IMU's accelerometers also can be processed to obtain velocity and displacement of the IMU.

[0024] The term “three-degrees of freedom tracking system” (3DOF tracking system) is used herein to refer to a device that tracks rotational movement. For example, the 3DOF tracking system can track whether a user of a head-wearable device is looking left or right, rotating their head up or down, and pivoting left or right. However, the head-wearable device cannot use the 3DOF tracking system to determine whether the user has moved around a scene by moving in the physical world. As such, 3DOF tracking system may not be accurate enough to be used for positional signals. The 3DOF tracking system may be part of an AR/VR display device that includes IMU sensors. For example, the 3DOF tracking system uses sensor data from sensors such as accelerometers, gyroscopes, and magnetometers.

[0025] The term “six-degrees of freedom tracking system” (6DOF tracking system) is used herein to refer to a device that tracks rotational and translational motion. For example, the 6DOF tracking system can track whether the user has rotated their head and moved forward or backward, laterally or vertically and up or down. The 6DOF tracking system may include a Simultaneous Localization and Mapping (SLAM) system and/or a VIO system that relies on data acquired from multiple sensors (e.g., depth cameras, inertial sensors). The 6DOF tracking system analyzes data from the sensors to accurately determine the pose of the display device.

[0026] The present application describes a system for translating sign language using collaborative augmented reality devices. For example, a first user wears a head-wearable AR device (e.g., smart glasses) and performs hand gesture (e.g., sign language). A second user facing the first user can also wear another AR device or can point a camera of his/her mobile communication device (e.g., smart phone) towards the hands of the first user. The cameras from the AR devices of the first and second users capture the hand gestures of the first user from different perspective or angles. The first AR device is first synchronized with the second AR device. For example, synchronization can be based on the timestamps of the images generated by the cameras of the first and second AR devices. In another example, synchronization can be based on the poses of the AR devices in same registered coordinate system. One or more processes of the sign language translation system is distributed among the first and second AR devices. For example, the first AR device can share images of the hand gesture captured with the camera of the first AR device to the second AR device. The second AR device may have more computational resources (e.g., a smart phone may have a faster processor, larger memory, and larger battery than smart glasses) and performs hand tracking, gesture recognition, and sign language translation. In one example, the distribution of the processes of the sign language translation system can be based on a comparison of the specifications of the first AR device and second AR device and other conditions (e.g., detection of occlusion of hands from camera of the first AR device; the system will rely on images from cameras of the second AR device when the hands are occluded in the cameras of the first AR device).

[0027] In other example embodiments, one AR device performs hand tracking and gesture recognition, and the other AR device performs gesture recognition. In other examples, results of the processes (sign language translation, hand skeleton detection, semantic gesture recognition) is shared between the AR devices temporally (e.g., every other frame) or spatially (e.g. left/right hand only). In another example, the first AR device displays suggested words/phrases according to the contextual information, user's history inputs, text corpus, pre-defined terms. Upon detection of a triggering condition (special gesture or actuation of the user input), the first user can select the suggested options to avoid the fingerspelling or speed up interpretation.

[0028] In one example embodiment, a method for recognizing sign language using collaborative AR devices is described. In one aspect, the method includes accessing a first image generated by a

first camera of a first augmented reality device and a second image generated by a second camera of a second augmented reality device, the first image and the second image depicting a hand gesture of a user of the first augmented reality device, synchronizing the first augmented reality device with the second augmented reality device, in response to synchronizing, distributing one or more processes of a sign language recognition system between the first augmented reality device and the second augmented reality device, where the one or more processes are performed on a corresponding augmented reality device, collecting results from the one or more processes from the first augmented reality device and from the second augmented reality device, and displaying, in near real-time in a first display of the first augmented reality device or in a second display of the second augmented reality device, text indicating a sign language translation of the hand gesture based on the results.

[0029] As a result, one or more of the methodologies described herein facilitate solving the technical problem of resource management of sign language translation. The presently described method provides an improvement to an operation of the functioning of a computer by distributing sign language processes among synchronized collaborative AR devices to provide power consumption reduction. As such, one or more of the methodologies described herein may obviate a need for certain efforts or computing resources. Examples of such computing resources include processor cycles, network traffic, memory usage, data storage capacity, power consumption, network bandwidth, and cooling capacity.

[0030] FIG. 1 is a block diagram illustrating a network environment **100** for sign language translation using collaborating AR devices in accordance with one example embodiment. The network environment **100** includes the AR device B **114**, the AR device A **110**, and the server **116**, communicatively coupled to each other via a network **104**. The AR device B **114**, AR device A **110**, and the server **116** may each be implemented in a computer system, in whole or in part, as described below with respect to FIG. 13. The server **116** may be part of a network-based system. For example, the network-based system may be or include a cloud-based server system that provides additional information, such as reference frame alignment data of the AR device B **114** and the AR device A **110**.

[0031] A user **106** wears the AR device A **110** (e.g., smart glasses) and operates the AR device A **110** via voice, hand gesture, physical input (e.g., touching a surface of the AR device A **110** or pressing on a button on the AR device A **110**). The user **106** performs hand gestures with his/her hands **108**. The AR device A **110** captures an image of the hands **108** using a camera (not shown) of the AR device A **110**. The AR device A **110** may be a computing device with a display. The computing device may be removable mounted to a head of the user **106**. In one example, the display may be a screen that displays what is captured with a camera of the AR device B **114**/AR device A **110**. In another example, the display of the device may be transparent, such as in lenses of wearable computing glasses, that allow the user **106** to view content presented on the display while simultaneously viewing real world object visible through the display.

[0032] A user **112** operates the AR device B **114**. The user **112** may be a human user (e.g., a human being), a machine user (e.g., a computer configured by a software program to interact with the AR device B **114**), or any suitable combination thereof (e.g., a human assisted by a machine or a machine supervised by a human). In one example, the user **112** aims a camera of the AR device B **114** at the hands **108** of the user **106** in the real world environment **102**. The AR device B **114** may be a computing device with a display such as a smartphone, a tablet computer, or a wearable computing device (e.g., watch or glasses). The computing device may be hand-held or may be removable mounted to a head of the user **112**). In one example, the display may be a screen that displays what is captured with a camera of the AR device B **114**/AR device A **110**. In another example, the display of the device may be transparent, such as in lenses of wearable computing glasses, that allow the user **112** to view content presented on the display while simultaneously viewing real world object visible through the display. In another example, the AR device B **114**

includes a computing device (e.g., a mobile device capable of displaying text or playing audio). [0033] In one example, the AR device A **110** and AR device B **114** each generate an image of the hands **108** from different viewpoints. As such, the images captured by the AR device B **114** and the AR device A **110** include overlapping regions.

[0034] The AR device A **110** includes a tracking system (not shown). The tracking system tracks the pose (e.g., position and orientation) of the AR device A **110** relative to the real world environment **102** using optical sensors (e.g., image camera), inertia sensors (e.g., gyroscope, accelerometer), wireless sensors (Bluetooth, Wi-Fi), GPS sensor, and audio sensor to determine the location of the AR device A **110** within the real world environment **102**. For example, the tracking system of the AR device A **110** includes a 6DOF tracking system that tracks whether the user **106** has rotated their head and moved forward or backward, laterally or vertically and up or down.

[0035] The AR device B **114** also includes a tracking system (not shown). The tracking system tracks the pose (e.g., position and orientation) of the AR device B **114** relative to the real world environment **102** using optical sensors (e.g., image camera), inertia sensors (e.g., gyroscope, accelerometer), wireless sensors (Bluetooth, Wi-Fi), GPS sensor, and audio sensor to determine the location of the AR device B **114** within the real world environment **102**. For example, the tracking system of the AR device B **114** includes a 6DOF tracking system that tracks whether the user **112** (in the case of a head-wearable device) has rotated their head and moved forward or backward, laterally or vertically and up or down.

[0036] In one example embodiment, the server **116** coordinates the distribution of sign language translation between the AR device B **114** and the AR device A **110**. For example, the server **116** identifies the available computational resources and technical specifications of the AR device B **114** and AR device A **110**, allocates one or more processes from the sign language translation to the AR device B **114** and AR device A **110** based on the available computational resources and technical specifications. In another example, the server **116** receives image data, image metadata (e.g., timestamp), and pose information from both the AR device B **114** and AR device A **110**, synchronizes the receiving information, performs the sign language translation based on the synchronized information, and provides a translation to both the AR device B **114** and the AR device A **110**. In another example, the server **116** provides a suggested translation to the AR device A **110** first, waits for the user **106** to confirm the suggested translation before sending the confirmed translation to the AR device B **114**.

[0037] Any of the machines, databases, or devices shown in FIG. **1** may be implemented in a general-purpose computer modified (e.g., configured or programmed) by software to be a special-purpose computer to perform one or more of the functions described herein for that machine, database, or device. For example, a computer system able to implement any one or more of the methodologies described herein is discussed below with respect to FIG. **9**. As used herein, a “database” is a data storage resource and may store data structured as a text file, a table, a spreadsheet, a relational database (e.g., an object-relational database), a triple store, a hierarchical data store, or any suitable combination thereof. Moreover, any two or more of the machines, databases, or devices illustrated in FIG. **1** may be combined into a single machine, and the functions described herein for any single machine, database, or device may be subdivided among multiple machines, databases, or devices.

[0038] The network **104** may be any network that enables communication between or among machines (e.g., server **116**), databases, and devices (e.g., AR device B **114**, AR device A **110**). Accordingly, the network **104** may be a wired network, a wireless network (e.g., a mobile or cellular network), or any suitable combination thereof. The network **104** may include one or more portions that constitute a private network, a public network (e.g., the Internet), or any suitable combination thereof.

[0039] FIG. **2** is a network diagram illustrating a network environment **200** suitable for operating AR device B **114** and AR device A **110**, according to some example embodiments. The network

environment **200** includes AR device B **114** and AR device A **110**. The AR device B **114** may communicate wirelessly (e.g., Bluetooth) with the AR device A **110**. In contrast to FIG. **1**, the AR device B **114** and AR device A **110** communicate directly with each other.

[0040] The AR device A **110** and the AR device B **114** may share data with each other. In one example, the AR device A **110** receives image data, pose data from the AR device B **114** and synchronizes the AR device A **110** with the AR device B **114** based on the received data from AR device B **114** and the image data and pose data of the AR device A **110**. In another example, the AR device B **114** receives image data, pose data from the AR device B **114** and synchronizes the AR device B **114** with the AR device A **110** based on the received data from AR device A **110** and the image data and pose data of the AR device B **114**.

[0041] FIG. **3** is a block diagram illustrating modules (e.g., components) of the AR device A **110**, according to some example embodiments. The AR device A **110** includes sensors **302**, a display **304**, a processor **308**, a Graphical processing unit **318**, a display controller **320**, and a storage device **306**. Examples of the AR device A **110** includes a wearable computing device. AR device B **114** includes components similar to AR device A **110**. However, AR device B **114** can include a wearable computing device, a tablet computer, or a smart phone.

[0042] The sensors **302** include an optical sensor **314** and an inertial sensor **316**. The optical sensor **314** includes stereo cameras. The inertial sensor **316** includes a combination of gyroscope, accelerometer, magnetometer. Other examples of sensors **302** include a proximity or location sensor (e.g., near field communication, GPS, Bluetooth, Wifi), an audio sensor (e.g., a microphone), or any suitable combination thereof. It is noted that the sensors **302** described herein are for illustration purposes and the sensors **302** are thus not limited to the ones described above. In one example embodiment, the sensors **302** can include a depth sensor such as a structured-light sensor, a time-of-flight sensor, passive stereo sensor, and an ultrasound device, time-of-flight sensor.

[0043] The display **304** includes a screen or monitor configured to display images generated by the processor **308**. In one example embodiment, the display **304** may be transparent or semi-transparent so that the user **106** can see through the display **304** (in AR use case). In another example, the display **304**, such as a LCOS display, presents each frame of virtual content in multiple presentations.

[0044] The processor **308** includes an AR application **310**, a 6DOF tracker **312**, a depth system **324**, and a collaborative sign language system **328**. The AR application **310** detects and identifies the hands **108** or an item in the real world environment **102** using computer vision. The AR application **310** retrieves a virtual object (e.g., 3D object model) based on an identified item (e.g., hands **108**) or physical environment. The display **304** displays the virtual object. The AR application **310** includes a local rendering engine that generates a visualization of a virtual object overlaid (e.g., superimposed upon, or otherwise displayed in tandem with) on an image of the item captured by the optical sensor **314**. A visualization of the virtual object may be manipulated by adjusting a position of the item (e.g., its physical location, orientation, or both) relative to the optical sensor **314**. Similarly, the visualization of the virtual object may be manipulated by adjusting a pose of the AR device A **110** relative to the item.

[0045] The 6DOF tracker **312** estimates a pose of the AR device A **110**. For example, the 6DOF tracker **312** uses image data and corresponding inertial data from the optical sensor **314** and the inertial sensor **316** to track a location and pose of the AR device A **110** relative to a frame of reference (e.g., real world environment **102**). In one example, the 6DOF tracker **312** uses the sensor data to determine the three-dimensional pose of the AR device A **110**. The three-dimensional pose is a determined orientation and position of the AR device A **110** in relation to the user's real world environment **102**. For example, the AR device A **110** may use images of the user's real world environment **102**, as well as other sensor data to identify a relative position and orientation of the AR device A **110** from physical objects in the real world environment **102** surrounding the AR

device A **110**. The 6DOF tracker **312** continually gathers and uses updated sensor data describing movements of the AR device A **110** to determine updated three-dimensional poses of the AR device A **110** that indicate changes in the relative position and orientation of the AR device A **110** from the physical objects in the real world environment **102**. The 6DOF tracker **312** provides the three-dimensional pose of the AR device A **110** to the depth system **324** and the collaborative sign language system **328**.

[0046] The depth system **324** accesses images from the optical sensor **314** and sparse 3D points from the 6DOF tracker **312** to predict depths and generate a dense point cloud. In one example embodiment, the optical sensor **314** includes a depth sensor or a stereo sensor.

[0047] The collaborative sign language system **328** coordinates the distribution of processes of a sign language translation application between the AR device B **114** and the AR device A **110** (and optionally the server **116**). For example, the collaborative sign language system **328** identifies the available computational resources and technical specifications of the AR device B **114** and AR device A **110**, allocates one or more processes from the sign language translation between the AR device B **114** and AR device A **110** based on the available computational resources and technical specifications. In another example, the AR device A **110** receives image data, image metadata (e.g., timestamp), and pose information from the AR device B **114**, synchronizes the AR device A **110** with the AR device B **114** using the receiving information, performs the sign language translation based on the synchronized information, displays the sign language translation in the display **304**, and provides the sign language translation to the AR device B **114**. In another example, the collaborative sign language system **328** displays a suggested translation in the display **304**, waits for the user **106** to confirm the suggested translation, detects the confirmation by detecting a predefined gesture that indicate a sign language suggestion confirmation gesture, and sends the confirmed translation to the AR device B **114**.

[0048] The Graphical processing unit **318** includes a render engine (not shown) that is configured to render a frame of a 3D model of a virtual object based on the virtual content provided by the AR application **310** and the pose of the AR device B **114** (relative to AR device A **110**). In other words, the Graphical processing unit **318** uses the three-dimensional pose of the AR device B **114** to generate frames of virtual content to be presented on the display **304**. For example, the Graphical processing unit **318** uses the three-dimensional pose to render a frame of the virtual content such that the virtual content is presented at an orientation and position in the display **304** to properly augment the user's reality. As an example, the Graphical processing unit **318** may use the three-dimensional pose data to render a frame of virtual content such that, when presented on the display **304**, the virtual content overlaps with a physical object in the user's real world environment **102**. The Graphical processing unit **318** generates updated frames of virtual content based on updated three-dimensional poses of the AR device B **114**, which reflect changes in the position and orientation of the user in relation to physical objects in the user's real world environment **102**.

[0049] The Graphical processing unit **318** transfers the rendered frame to the display controller **320**. The display controller **320** is positioned as an intermediary between the Graphical processing unit **318** and the display **304**, receives the image data (e.g., rendered frame) from the Graphical processing unit **318**, provides the rendered frame to display **304**.

[0050] The storage device **306** stores virtual object content **322** and gesture data **326** (e.g., predefined gesture indicating suggested word confirmation). The virtual object content **322** includes, for example, a database of visual references (e.g., images, QR codes) and corresponding virtual content (e.g., three-dimensional model of virtual objects). The gesture data **326** indicate predefined hand gestures or predefined user input to indicate a confirmation of suggested words translated by the collaborative sign language system **328** or by another translation system (located on the AR device B **114** or the server **116**). The gesture data **326** can also define other triggering conditions that when detected enable the user **106** to select the suggested options to avoid the fingerspelling or speed up sign language interpretation.

[0051] Any one or more of the modules described herein may be implemented using hardware (e.g., a Processor of a machine) or a combination of hardware and software. For example, any module described herein may configure a processor to perform the operations described herein for that module. Moreover, any two or more of these modules may be combined into a single module, and the functions described herein for a single module may be subdivided among multiple modules. Furthermore, according to various example embodiments, modules described herein as being implemented within a single machine, database, or device may be distributed across multiple machines, databases, or devices.

[0052] FIG. 4 is a block diagram illustrating the collaborative sign language system 328 in accordance with one example embodiment. The collaborative sign language system 328 includes an AR device interface 402, a synchronization module 404, a hand tracking application 406, and a sign language recognition application 408.

[0053] The AR device interface 402 communicates with the AR device B 114. In one example, the AR device interface 402 accesses 6DoF pose data and camera data from the AR device B 114. In another example, the AR device interface 402 shares the 6DoF pose data of 6DOF tracker 312 and camera data of the optical sensor 314 to the AR device B 114. In another example, the AR device interface 402 communicates which processes the AR device B 114 is to perform as part of the collaborative sign language translation process.

[0054] The synchronization module 404 accesses the 6DOF pose data from 6DOF tracker 312 and the camera data from optical sensor 314, the 6DoF pose data and camera data (including timestamp metadata) from the AR device B 114. In another example, the synchronization module 404 accesses the technical specifications and available computation resources of AR device A 110 and AR device B 114. The synchronization module 404 synchronizes the AR device A 110 with the AR device B 114 by matching the timestamps of images from the AR device A 110 with images from the AR device B 114. In another example, the synchronization can be performed based on aligning/registering the 6DoF poses of the cameras of the AR device A 110 and the AR device B 114 in a same common coordinate system.

[0055] The hand tracking application 406 detects the hands 108 using computer vision. In one example, the hand tracking application 406 identifies features of the hands (motion, skeleton, depth) and tracks the motion of hands 108. The hand tracking application 406 can include a number of processes, such as for example, hand tracking process, hand gesture detection process, and sign language translation process.

[0056] The sign language recognition application 408 identifies a gesture of the hands 108 based on the hand tracking application 406 and identifies a word corresponding to the gesture. In one example, the sign language recognition application 408 includes a sign language translation process. In another example embodiment, the sign language recognition application 408 identifies a suggested word based on a combination of contextual information of the AR device A 110 (e.g., profile of the user 106, geo-location of the AR device A 110), historical inputs of the user 106, text corpus, and pre-defined terms). The sign language recognition application 408 provides text (indicating the suggested word) to the display 304. The display 304 displays the text. In another example, the sign language recognition application 408 communicates the text to the AR device B 114 where the text is displayed in a screen of the AR device B 114.

[0057] FIG. 5 is an interaction diagram illustrating interactions between collaborative augmented reality devices (e.g., AR device A 110 and AR device B 114) in accordance with one example embodiment. The AR device B 114 captures images of the hands 108 with its camera. The AR device A 110 captures images of hands 108 with its camera. At 502, the AR device B 114 sends the images, camera data, and 6DoF pose data to the AR device A 110. At 504, the AR device A 110 synchronizes the AR device A 110 and the AR device B 114 based on the image data from both AR device A 110 and AR device B 114. At 506, the AR device A 110 applies a hand tracking process based on the synchronized images (e.g., a combination of image data from AR device A 110 and

AR device B **114**). At **508**, the AR device A **110** applies a gesture recognition process to the synchronized images to generate sign language translation. At **510**, the AR device A **110** displays the sign language translation in the display **304**, and provides the sign language translation to the AR device B **114**.

[0058] FIG. **6** is an interaction diagram illustrating interactions between collaborative augmented reality devices (e.g., AR device A **110** and AR device B **114**) in accordance with one example embodiment. The AR device B **114** captures images of the hands **108** with its camera. The AR device A **110** captures images of hands **108** with its camera. At **602**, the AR device B **114** sends the images, camera data, and 6DoF pose data to the AR device A **110**. At **604**, the AR device A **110** synchronizes the AR device A **110** and the AR device B **114** based on the image data from both AR device A **110** and AR device B **114**. At **606**, the AR device A **110** applies a hand tracking process based on the synchronized images (e.g., a combination of image data from AR device A **110** and AR device B **114**). At **608**, the AR device A **110** applies a gesture recognition process to the synchronized images to generate a suggested sign language translation. At **610**, the AR device A **110** displays the suggested translation to the user **106** in the display **304**. At **612**, the AR device A **110** detects a confirmation of the suggested translation to the user **106**. At **614**, the AR device A **110** shares the translation to the AR device B **114**.

[0059] FIG. **7** is an interaction diagram illustrating interactions between collaborative augmented reality devices (e.g., AR device A **110** and AR device B **114**) in accordance with one example embodiment. The AR device B **114** captures images of the hands **108** with its camera. The AR device A **110** captures images of hands **108** with its camera. At **702**, the AR device B **114** sends the images, camera data, and 6DoF pose data to the AR device A **110**. At **704**, the AR device A **110** synchronizes the AR device A **110** and the AR device B **114** based on the image data from both AR device A **110** and AR device B **114**. At **706**, the AR device A **110** applies a hand tracking process based on the synchronized images (e.g., a combination of image data from AR device A **110** and AR device B **114**). The AR device A **110** sends the results of the hand tracking process to the AR device B **114**. At **708**, the AR device B **114** applies a gesture recognition process to the results of the hand tracking processed at AR device A **110**. At **710**, the AR device B **114** applies a sign language recognition process on the results of the gesture recognition process. At **712**, the AR device B **114** displays the sign language translation (e.g., text) in a display of the AR device B **114**. At **714**, the AR device B **114** provides the text to AR device A **110**. At **716**, the AR device A **110** display the text.

[0060] FIG. **8** is an interaction diagram illustrating interactions between collaborative augmented reality devices (e.g., AR device A **110** and AR device B **114**) in accordance with one example embodiment. The AR device B **114** captures images of the hands **108** with its camera. The AR device A **110** captures images of hands **108** with its camera. At **802**, the AR device B **114** sends the images, camera data, and 6DoF pose data to the AR device A **110**. At **804**, the AR device A **110** synchronizes the AR device A **110** and the AR device B **114** based on the image data from both AR device A **110** and AR device B **114**. At **806**, the AR device A **110** provides an egocentric view image to the AR device B **114**. At **808**, the AR device A **110** applies a hand tracking process based on the synchronized images (e.g., a combination of image data from AR device B **114** and the egocentric view data from AR device A **110**). At **810**, the AR device B **114** applies a gesture recognition process and sign language recognition process to the results of the hand tracking process. At **812**, the AR device B **114** displays the sign language translation (e.g., text) in a display of the AR device B **114**. At **814**, the AR device B **114** provides the text to AR device A **110** for display.

[0061] FIG. **9** is a flow diagram illustrating a method **900** for sign language translation in accordance with one example embodiment. Operations in the method **900** may be performed by the collaborative sign language system **328**, using components (e.g., modules, engines) described above with respect to FIG. **4**. Accordingly, the method **900** is described by way of example with

reference to the collaborative sign language system **328**. However, it shall be appreciated that at least some of the operations of the method **900** may be deployed on various other hardware configurations or be performed by similar components residing elsewhere.

[0062] In block **902**, the synchronization module **404** synchronizes a signer's AR device (e.g., AR device A **110**) with a recipient's AR device (e.g., AR device B **114**). In block **904**, the hand tracking application **406** performs hand tracking and the sign language recognition application **408** performs sign language recognition on the synchronized images. In block **906**, the display **304** displays the suggested translation based on sign language recognition on the signer's AR device. In block **908**, the sign language recognition application **408** detects confirmation of suggested translation on the signer's AR device. In block **910**, the sign language recognition application **408** provides the suggested translation to the recipient's AR device.

[0063] It is to be noted that other embodiments may use different sequencing, additional or fewer operations, and different nomenclature or terminology to accomplish similar functions. In some embodiments, various operations may be performed in parallel with other operations, either in a synchronous or asynchronous manner. The operations described herein were chosen to illustrate some principles of operations in a simplified form.

[0064] FIG. **10** is a flow diagram illustrating a method for distributing sign language translation processes in accordance with one example embodiment. Operations in the method **1000** may be performed by the collaborative sign language system **328**, using components (e.g., modules, engines) described above with respect to FIG. **4**. Accordingly, the method **1000** is described by way of example with reference to the collaborative sign language system **328**. However, it shall be appreciated that at least some of the operations of the method **1000** may be deployed on various other hardware configurations or be performed by similar components residing elsewhere.

[0065] In block **1002**, the collaborative sign language system **328** synchronizes a signer's AR device with a recipient's AR device. In block **1004**, the collaborative sign language system **328** distributes one or more processes from hand tracking, gesture recognition, and sign language recognition between the signer's AR device and the recipient's AR device. In block **1006**, the collaborative sign language system **328** shares results of processes between signer's AR device and recipient's AR device. In block **1008**, the display **304** displays translation based on shared results.

[0066] It is to be noted that other embodiments may use different sequencing, additional or fewer operations, and different nomenclature or terminology to accomplish similar functions. In some embodiments, various operations may be performed in parallel with other operations, either in a synchronous or asynchronous manner. The operations described herein were chosen to illustrate some principles of operations in a simplified form.

System with Head-Wearable Apparatus

[0067] FIG. **11** illustrates a network environment **1100** in which the head-wearable apparatus **1102** can be implemented according to one example embodiment. FIG. **11** is a high-level functional block diagram of an example head-wearable apparatus **1102** communicatively coupled a mobile client device **1138** and a server system **1132** via various network **1140**.

[0068] head-wearable apparatus **1102** includes a camera, such as at least one of visible light camera **1112**, infrared emitter **1114** and infrared camera **1116**. The client device **1138** can be capable of connecting with head-wearable apparatus **1102** using both a communication **1134** and a communication **1136**. client device **1138** is connected to server system **1132** and network **1140**. The network **1140** may include any combination of wired and wireless connections.

[0069] The head-wearable apparatus **1102** further includes two image displays of the image display of optical assembly **1104**. The two include one associated with the left lateral side and one associated with the right lateral side of the head-wearable apparatus **1102**. The head-wearable apparatus **1102** also includes image display driver **1108**, image processor **1110**, low-power low power circuitry **1126**, and high-speed circuitry **1118**. The image display of optical assembly **1104** are for presenting images and videos, including an image that can include a graphical user interface

to a user of the head-wearable apparatus **1102**.

[0070] The image display driver **1108** commands and controls the image display of the image display of optical assembly **1104**. The image display driver **1108** may deliver image data directly to the image display of the image display of optical assembly **1104** for presentation or may have to convert the image data into a signal or data format suitable for delivery to the image display device. For example, the image data may be video data formatted according to compression formats, such as H.264 (MPEG-4 Part 10), HEVC, Theora, Dirac, RealVideo RV40, VP8, VP9, or the like, and still image data may be formatted according to compression formats such as Portable Network Group (PNG), Joint Photographic Experts Group (JPEG), Tagged Image File Format (TIFF) or exchangeable image file format (Exif) or the like.

[0071] As noted above, head-wearable apparatus **1102** includes a frame and stems (or temples) extending from a lateral side of the frame. The head-wearable apparatus **1102** further includes a user input device **1106** (e.g., touch sensor or push button) including an input surface on the head-wearable apparatus **1102**. The user input device **1106** (e.g., touch sensor or push button) is to receive from the user an input selection to manipulate the graphical user interface of the presented image.

[0072] The components shown in FIG. **11** for the head-wearable apparatus **1102** are located on one or more circuit boards, for example a PCB or flexible PCB, in the rims or temples. Alternatively, or additionally, the depicted components can be located in the chunks, frames, hinges, or bridge of the head-wearable apparatus **1102**. Left and right can include digital camera elements such as a complementary metal-oxide-semiconductor (CMOS) image sensor, charge coupled device, a camera lens, or any other respective visible or light capturing elements that may be used to capture data, including images of scenes with unknown objects.

[0073] The head-wearable apparatus **1102** includes a memory **1122** which stores instructions to perform a subset or all of the functions described herein. memory **1122** can also include storage device.

[0074] As shown in FIG. **11**, high-speed circuitry **1118** includes high-speed processor **1120**, memory **1122**, and high-speed wireless circuitry **1124**. In the example, the image display driver **1108** is coupled to the high-speed circuitry **1118** and operated by the high-speed processor **1120** in order to drive the left and right image displays of the image display of optical assembly **1104**. high-speed processor **1120** may be any processor capable of managing high-speed communications and operation of any general computing system needed for head-wearable apparatus **1102**. The high-speed processor **1120** includes processing resources needed for managing high-speed data transfers on communication **1136** to a wireless local area network (WLAN) using high-speed wireless circuitry **1124**. In certain examples, the high-speed processor **1120** executes an operating system such as a LINUX operating system or other such operating system of the head-wearable apparatus **1102** and the operating system is stored in memory **1122** for execution. In addition to any other responsibilities, the high-speed processor **1120** executing a software architecture for the head-wearable apparatus **1102** is used to manage data transfers with high-speed wireless circuitry **1124**. In certain examples, high-speed wireless circuitry **1124** is configured to implement Institute of Electrical and Electronic Engineers (IEEE) 1102.11 communication standards, also referred to herein as Wi-Fi. In other examples, other high-speed communications standards may be implemented by high-speed wireless circuitry **1124**.

[0075] The low power wireless circuitry **1130** and the high-speed wireless circuitry **1124** of the head-wearable apparatus **1102** can include short range transceivers (Bluetooth™) and wireless wide, local, or wide area network transceivers (e.g., cellular or WiFi). The client device **1138**, including the transceivers communicating via the communication **1134** and communication **1136**, may be implemented using details of the architecture of the head-wearable apparatus **1102**, as can other elements of network **1140**.

[0076] The memory **1122** includes any storage device capable of storing various data and

applications, including, among other things, camera data generated by the left and right, infrared camera **1116**, and the image processor **1110**, as well as images generated for display by the image display driver **1108** on the image displays of the image display of optical assembly **1104**. While memory **1122** is shown as integrated with high-speed circuitry **1118**, in other examples, memory **1122** may be an independent standalone element of the head-wearable apparatus **1102**. In certain such examples, electrical routing lines may provide a connection through a chip that includes the high-speed processor **1120** from the image processor **1110** or low power processor **1128** to the memory **1122**. In other examples, the high-speed processor **1120** may manage addressing of memory **1122** such that the low power processor **1128** will boot the high-speed processor **1120** any time that a read or write operation involving memory **1122** is needed.

[0077] As shown in FIG. **11**, the low power processor **1128** or high-speed processor **1120** of the head-wearable apparatus **1102** can be coupled to the camera (visible light camera **1112**; infrared emitter **1114**, or infrared camera **1116**), the image display driver **1108**, the user input device **1106** (e.g., touch sensor or push button), and the memory **1122**.

[0078] The head-wearable apparatus **1102** is connected with a host computer. For example, the head-wearable apparatus **1102** is paired with the client device **1138** via the communication **1136** or connected to the server system **1132** via the network **1140**. server system **1132** may be one or more computing devices as part of a service or network computing system, for example, that include a processor, a memory, and network communication interface to communicate over the network **1140** with the client device **1138** and head-wearable apparatus **1102**.

[0079] The client device **1138** includes a processor and a network communication interface coupled to the processor. The network communication interface allows for communication over the network **1140**, communication **1134** or communication **1136**. client device **1138** can further store at least portions of the instructions for generating a binaural audio content in the client device **1138**'s memory to implement the functionality described herein.

[0080] Output components of the head-wearable apparatus **1102** include visual components, such as a display such as a liquid crystal display (LCD), a plasma display panel (PDP), a light emitting diode (LED) display, a projector, or a waveguide. The image displays of the optical assembly are driven by the image display driver **1108**. The output components of the head-wearable apparatus **1102** further include acoustic components (e.g., speakers), haptic components (e.g., a vibratory motor), other signal generators, and so forth. The input components of the head-wearable apparatus **1102**, the client device **1138**, and server system **1132**, such as the user input device **1106**, may include alphanumeric input components (e.g., a keyboard, a touch screen configured to receive alphanumeric input, a photo-optical keyboard, or other alphanumeric input components), point-based input components (e.g., a mouse, a touchpad, a trackball, a joystick, a motion sensor, or other pointing instruments), tactile input components (e.g., a physical button, a touch screen that provides location and force of touches or touch gestures, or other tactile input components), audio input components (e.g., a microphone), and the like.

[0081] The head-wearable apparatus **1102** may optionally include additional peripheral device elements. Such peripheral device elements may include biometric sensors, additional sensors, or display elements integrated with head-wearable apparatus **1102**. For example, peripheral device elements may include any I/O components including output components, motion components, position components, or any other such elements described herein.

[0082] For example, the biometric components include components to detect expressions (e.g., hand expressions, facial expressions, vocal expressions, body gestures, or eye tracking), measure biosignals (e.g., blood pressure, heart rate, body temperature, perspiration, or brain waves), identify a person (e.g., voice identification, retinal identification, facial identification, fingerprint identification, or electroencephalogram based identification), and the like. The motion components include acceleration sensor components (e.g., accelerometer), gravitation sensor components, rotation sensor components (e.g., gyroscope), and so forth. The position components include

location sensor components to generate location coordinates (e.g., a Global Positioning System (GPS) receiver component), WiFi or Bluetooth™ transceivers to generate positioning system coordinates, altitude sensor components (e.g., altimeters or barometers that detect air pressure from which altitude may be derived), orientation sensor components (e.g., magnetometers), and the like. Such positioning system coordinates can also be received over and communication **1136** from the client device **1138** via the low power wireless circuitry **1130** or high-speed wireless circuitry **1124**. [0083] Where a phrase similar to “at least one of A, B, or C,” “at least one of A, B, and C,” “one or more A, B, or C,” or “one or more of A, B, and C” is used, it is intended that the phrase be interpreted to mean that A alone may be present in an embodiment, B alone may be present in an embodiment, C alone may be present in an embodiment, or that any combination of the elements A, B and C may be present in a single embodiment; for example, A and B, A and C, B and C, or A and B and C.

[0084] Changes and modifications may be made to the disclosed embodiments without departing from the scope of the present disclosure. These and other changes or modifications are intended to be included within the scope of the present disclosure, as expressed in the following claims.

[0085] FIG. **12** is a block diagram **1200** illustrating a software architecture **1204**, which can be installed on any one or more of the devices described herein. The software architecture **1204** is supported by hardware such as a machine **1202** that includes Processors **1220**, memory **1226**, and I/O Components **1238**. In this example, the software architecture **1204** can be conceptualized as a stack of layers, where each layer provides a particular functionality. The software architecture **1204** includes layers such as an operating system **1212**, libraries **1210**, frameworks **1208**, and applications **1206**. Operationally, the applications **1206** invoke API calls **1250** through the software stack and receive messages **1252** in response to the API calls **1250**.

[0086] The operating system **1212** manages hardware resources and provides common services. The operating system **1212** includes, for example, a kernel **1214**, services **1216**, and drivers **1222**. The kernel **1214** acts as an abstraction layer between the hardware and the other software layers. For example, the kernel **1214** provides memory management, Processor management (e.g., scheduling), Component management, networking, and security settings, among other functionality. The services **1216** can provide other common services for the other software layers. The drivers **1222** are responsible for controlling or interfacing with the underlying hardware. For instance, the drivers **1222** can include display drivers, camera drivers, BLUETOOTH® or BLUETOOTH® Low Energy drivers, flash memory drivers, serial communication drivers (e.g., Universal Serial Bus (USB) drivers), WI-FI® drivers, audio drivers, power management drivers, and so forth.

[0087] The libraries **1210** provide a low-level common infrastructure used by the applications **1206**. The libraries **1210** can include system libraries **1218** (e.g., C standard library) that provide functions such as memory allocation functions, string manipulation functions, mathematic functions, and the like. In addition, the libraries **1210** can include API libraries **1224** such as media libraries (e.g., libraries to support presentation and manipulation of various media formats such as Moving Picture Experts Group-4 (MPEG4), Advanced Video Coding (H.264 or AVC), Moving Picture Experts Group Layer-3 (MP3), Advanced Audio Coding (AAC), Adaptive Multi-Rate (AMR) audio codec, Joint Photographic Experts Group (JPEG or JPG), or Portable Network Graphics (PNG)), graphics libraries (e.g., an OpenGL framework used to render in two dimensions (2D) and three dimensions (3D) in a graphic content on a display), database libraries (e.g., SQLite to provide various relational database functions), web libraries (e.g., WebKit to provide web browsing functionality), and the like. The libraries **1210** can also include a wide variety of other libraries **1228** to provide many other APIs to the applications **1206**.

[0088] The frameworks **1208** provide a high-level common infrastructure that is used by the applications **1206**. For example, the frameworks **1208** provide various graphical user interface (GUI) functions, high-level resource management, and high-level location services. The

frameworks **1208** can provide a broad spectrum of other APIs that can be used by the applications **1206**, some of which may be specific to a particular operating system or platform.

[0089] In an example embodiment, the applications **1206** may include a home application **1236**, a contacts application **1230**, a browser application **1232**, a book reader application **1234**, a location application **1242**, a media application **1244**, a messaging application **1246**, a game application **1248**, and a broad assortment of other applications such as a third-party application **1240**. The applications **1206** are programs that execute functions defined in the programs. Various programming languages can be employed to create one or more of the applications **1206**, structured in a variety of manners, such as object-oriented programming languages (e.g., Objective-C, Java, or C++) or procedural programming languages (e.g., C or assembly language). In a specific example, the third-party application **1240** (e.g., an application developed using the ANDROID™ or IOS™ software development kit (SDK) by an entity other than the vendor of the particular platform) may be mobile software running on a mobile operating system such as IOS™, ANDROID™, WINDOWS® Phone, or another mobile operating system. In this example, the third-party application **1240** can invoke the API calls **1250** provided by the operating system **1212** to facilitate functionality described herein.

[0090] FIG. **13** is a diagrammatic representation of the machine **1300** within which instructions **1308** (e.g., software, a program, an application, an applet, an app, or other executable code) for causing the machine **1300** to perform any one or more of the methodologies discussed herein may be executed. For example, the instructions **1308** may cause the machine **1300** to execute any one or more of the methods described herein. The instructions **1308** transform the general, non-programmed machine **1300** into a particular machine **1300** programmed to carry out the described and illustrated functions in the manner described. The machine **1300** may operate as a standalone device or may be coupled (e.g., networked) to other machines. In a networked deployment, the machine **1300** may operate in the capacity of a server machine or a client machine in a server-client network environment, or as a peer machine in a peer-to-peer (or distributed) network environment. The machine **1300** may comprise, but not be limited to, a server computer, a client computer, a personal computer (PC), a tablet computer, a laptop computer, a netbook, a set-top box (STB), a PDA, an entertainment media system, a cellular telephone, a smart phone, a mobile device, a wearable device (e.g., a smart watch), a smart home device (e.g., a smart appliance), other smart devices, a web appliance, a network router, a network switch, a network bridge, or any machine capable of executing the instructions **1308**, sequentially or otherwise, that specify actions to be taken by the machine **1300**. Further, while only a single machine **1300** is illustrated, the term “machine” shall also be taken to include a collection of machines that individually or jointly execute the instructions **1308** to perform any one or more of the methodologies discussed herein.

[0091] The machine **1300** may include Processors **1302**, memory **1304**, and I/O Components **1342**, which may be configured to communicate with each other via a bus **1344**. In an example embodiment, the Processors **1302** (e.g., a Central Processing Unit (CPU), a Reduced Instruction Set Computing (RISC) Processor, a Complex Instruction Set Computing (CISC) Processor, a Graphics Processing Unit (GPU), a Digital Signal Processor (DSP), an ASIC, a Radio-Frequency Integrated Circuit (RFIC), another Processor, or any suitable combination thereof) may include, for example, a Processor **1306** and a Processor **1310** that execute the instructions **1308**. The term “Processor” is intended to include multi-core Processors that may comprise two or more independent Processors (sometimes referred to as “cores”) that may execute instructions contemporaneously. Although FIG. **13** shows multiple Processors **1302**, the machine **1300** may include a single Processor with a single core, a single Processor with multiple cores (e.g., a multi-core Processor), multiple Processors with a single core, multiple Processors with multiples cores, or any combination thereof.

[0092] The memory **1304** includes a main memory **1312**, a static memory **1314**, and a storage unit **1316**, both accessible to the Processors **1302** via the bus **1344**. The main memory **1304**, the static

memory **1314**, and storage unit **1316** store the instructions **1308** embodying any one or more of the methodologies or functions described herein. The instructions **1308** may also reside, completely or partially, within the main memory **1312**, within the static memory **1314**, within machine-readable medium **1318** within the storage unit **1316**, within at least one of the Processors **1302** (e.g., within the Processor's cache memory), or any suitable combination thereof, during execution thereof by the machine **1300**.

[0093] The I/O Components **1342** may include a wide variety of Components to receive input, provide output, produce output, transmit information, exchange information, capture measurements, and so on. The specific I/O Components **1342** that are included in a particular machine will depend on the type of machine. For example, portable machines such as mobile phones may include a touch input device or other such input mechanisms, while a headless server machine will likely not include such a touch input device. It will be appreciated that the I/O Components **1342** may include many other Components that are not shown in FIG. **13**. In various example embodiments, the I/O Components **1342** may include output Components **1328** and input Components **1330**. The output Components **1328** may include visual Components (e.g., a display such as a plasma display panel (PDP), a light emitting diode (LED) display, a liquid crystal display (LCD), a projector, or a cathode ray tube (CRT)), acoustic Components (e.g., speakers), haptic Components (e.g., a vibratory motor, resistance mechanisms), other signal generators, and so forth. The input Components **1330** may include alphanumeric input Components (e.g., a keyboard, a touch screen configured to receive alphanumeric input, a photo-optical keyboard, or other alphanumeric input Components), point-based input Components (e.g., a mouse, a touchpad, a trackball, a joystick, a motion sensor, or another pointing instrument), tactile input Components (e.g., a physical button, a touch screen that provides location and/or force of touches or touch gestures, or other tactile input Components), audio input Components (e.g., a microphone), and the like.

[0094] In further example embodiments, the I/O Components **1342** may include biometric Components **1332**, motion Components **1334**, environmental Components **1336**, or position Components **1338**, among a wide array of other Components. For example, the biometric Components **1332** include Components to detect expressions (e.g., hand expressions, facial expressions, vocal expressions, body gestures, or eye tracking), measure biosignals (e.g., blood pressure, heart rate, body temperature, perspiration, or brain waves), identify a person (e.g., voice identification, retinal identification, facial identification, fingerprint identification, or electroencephalogram-based identification), and the like. The motion Components **1334** include acceleration sensor Components (e.g., accelerometer), gravitation sensor Components, rotation sensor Components (e.g., gyroscope), and so forth. The environmental Components **1336** include, for example, illumination sensor Components (e.g., photometer), temperature sensor Components (e.g., one or more thermometers that detect ambient temperature), humidity sensor Components, pressure sensor Components (e.g., barometer), acoustic sensor Components (e.g., one or more microphones that detect background noise), proximity sensor Components (e.g., infrared sensors that detect nearby objects), gas sensors (e.g., gas detection sensors to detection concentrations of hazardous gases for safety or to measure pollutants in the atmosphere), or other Components that may provide indications, measurements, or signals corresponding to a surrounding physical environment. The position Components **1338** include location sensor Components (e.g., a GPS receiver Component), altitude sensor Components (e.g., altimeters or barometers that detect air pressure from which altitude may be derived), orientation sensor Components (e.g., magnetometers), and the like.

[0095] Communication may be implemented using a wide variety of technologies. The I/O Components **1342** further include communication Components **1340** operable to couple the machine **1300** to a network **1320** or devices **1322** via a coupling **1324** and a coupling **1326**, respectively. For example, the communication Components **1340** may include a network interface

Component or another suitable device to interface with the network **1320**. In further examples, the communication Components **1340** may include wired communication Components, wireless communication Components, cellular communication Components, Near Field Communication (NFC) Components, Bluetooth® Components (e.g., Bluetooth® Low Energy), Wi-Fi® Components, and other communication Components to provide communication via other modalities. The devices **1322** may be another machine or any of a wide variety of peripheral devices (e.g., a peripheral device coupled via a USB).

[0096] Moreover, the communication Components **1340** may detect identifiers or include Components operable to detect identifiers. For example, the communication Components **1340** may include Radio Frequency Identification (RFID) tag reader Components, NFC smart tag detection Components, optical reader Components (e.g., an optical sensor to detect one-dimensional bar codes such as Universal Product Code (UPC) bar code, multi-dimensional bar codes such as Quick Response (QR) code, Aztec code, Data Matrix, Dataglyph, MaxiCode, PDF417, Ultra Code, UCC RSS-2D bar code, and other optical codes), or acoustic detection Components (e.g., microphones to identify tagged audio signals). In addition, a variety of information may be derived via the communication Components **1340**, such as location via Internet Protocol (IP) geolocation, location via Wi-Fi® signal triangulation, location via detecting an NFC beacon signal that may indicate a particular location, and so forth.

[0097] The various memories (e.g., memory **1304**, main memory **1312**, static memory **1314**, and/or memory of the Processors **1302**) and/or storage unit **1316** may store one or more sets of instructions and data structures (e.g., software) embodying or used by any one or more of the methodologies or functions described herein. These instructions (e.g., the instructions **1308**), when executed by Processors **1302**, cause various operations to implement the disclosed embodiments.

[0098] The instructions **1308** may be transmitted or received over the network **1320**, using a transmission medium, via a network interface device (e.g., a network interface Component included in the communication Components **1340**) and using any one of a number of well-known transfer protocols (e.g., hypertext transfer protocol (HTTP)). Similarly, the instructions **1308** may be transmitted or received using a transmission medium via the coupling **1326** (e.g., a peer-to-peer coupling) to the devices **1322**.

[0099] As used herein, the terms “Machine-Storage Medium,” “device-storage medium,” and “computer-storage medium” mean the same thing and may be used interchangeably in this disclosure. The terms refer to a single or multiple storage devices and/or media (e.g., a centralized or distributed database, and/or associated caches and servers) that store executable instructions and/or data. The terms shall accordingly be taken to include, but not be limited to, solid-state memories, and optical and magnetic media, including memory internal or external to processors. Specific examples of Machine-Storage Media, computer-storage media, and/or device-storage media include non-volatile memory, including by way of example semiconductor memory devices, e.g., erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), field-programmable gate arrays (FPGAs), and flash memory devices; magnetic disks such as internal hard disks and removable disks; magneto-optical disks; and CD-ROM and DVD-ROM disks. The terms “Machine-Storage Media,” “computer-storage media,” and “device-storage media” specifically exclude carrier waves, modulated data signals, and other such media, at least some of which are covered under the term “signal medium” discussed below.

[0100] The terms “transmission medium” and “signal medium” mean the same thing and may be used interchangeably in this disclosure. The terms “transmission medium” and “signal medium” shall be taken to include any intangible medium that is capable of storing, encoding, or carrying the instructions **1416** for execution by the machine **1400**, and include digital or analog communications signals or other intangible media to facilitate communication of such software. Hence, the terms “transmission medium” and “signal medium” shall be taken to include any form of modulated data

signal, carrier wave, and so forth. The term “modulated data signal” means a signal that has one or more of its characteristics set or changed in such a manner as to encode information in the signal. [0101] The terms “machine-readable medium,” “Computer-Readable Medium,” and “device-readable medium” mean the same thing and may be used interchangeably in this disclosure. The terms are defined to include both Machine-Storage Media and transmission media. Thus, the terms include both storage devices/media and carrier waves/modulated data signals.

[0102] Although an embodiment has been described with reference to specific example embodiments, it will be evident that various modifications and changes may be made to these embodiments without departing from the broader scope of the present disclosure. Accordingly, the specification and drawings are to be regarded in an illustrative rather than a restrictive sense. The accompanying drawings that form a part hereof, show by way of illustration, and not of limitation, specific embodiments in which the subject matter may be practiced. The embodiments illustrated are described in sufficient detail to enable those skilled in the art to practice the teachings disclosed herein. Other embodiments may be utilized and derived therefrom, such that structural and logical substitutions and changes may be made without departing from the scope of this disclosure. This Detailed Description, therefore, is not to be taken in a limiting sense, and the scope of various embodiments is defined only by the appended claims, along with the full range of equivalents to which such claims are entitled.

[0103] Such embodiments of the inventive subject matter may be referred to herein, individually and/or collectively, by the term “invention” merely for convenience and without intending to voluntarily limit the scope of this application to any single invention or inventive concept if more than one is in fact disclosed. Thus, although specific embodiments have been illustrated and described herein, it should be appreciated that any arrangement calculated to achieve the same purpose may be substituted for the specific embodiments shown. This disclosure is intended to cover any and all adaptations or variations of various embodiments. Combinations of the above embodiments, and other embodiments not specifically described herein, will be apparent to those of skill in the art upon reviewing the above description.

[0104] The Abstract of the Disclosure is provided to allow the reader to quickly ascertain the nature of the technical disclosure. It is submitted with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. In addition, in the foregoing Detailed Description, it can be seen that various features are grouped together in a single embodiment for the purpose of streamlining the disclosure. This method of disclosure is not to be interpreted as reflecting an intention that the claimed embodiments require more features than are expressly recited in each claim. Rather, as the following claims reflect, inventive subject matter lies in less than all features of a single disclosed embodiment. Thus the following claims are hereby incorporated into the Detailed Description, with each claim standing on its own as a separate embodiment.

EXAMPLES

[0105] Example 1 is a method comprising: accessing a first image generated by a first camera of a first augmented reality device and a second image generated by a second camera of a second augmented reality device, the first image and the second image depicting a hand gesture of a user of the first augmented reality device; synchronizing the first augmented reality device with the second augmented reality device; in response to synchronizing, distributing one or more processes of a sign language recognition system between the first augmented reality device and the second augmented reality device, wherein the one or more processes are performed on a corresponding augmented reality device; collecting results from the one or more processes from the first augmented reality device and from the second augmented reality device; and displaying, in near real-time in a first display of the first augmented reality device or in a second display of the second augmented reality device, text indicating a sign language translation of the hand gesture based on the results.

[0106] Example 2 includes the method of example 1, wherein synchronizing the first augmented reality device with the second augmented reality device further comprises: mapping a first timestamp of the first image with a second timestamp of the second image.

[0107] Example 3 includes the method of example 1, wherein synchronizing the first augmented reality device with the second augmented reality device further comprises: registering a six-degree-of-freedom coordinate system of the first augmented reality device with a six-degree-of-freedom coordinate system of the second augmented reality device.

[0108] Example 4 includes the method of example 1, further comprising: communicating the first image to the second augmented reality device, the first image providing an egocentric view of the hand gesture of the user of the first augmented reality device, and wherein the second augmented reality device is configured to perform the one or more processes of the sign language recognition system on a combination of the first image, a first hand skeleton based on the first image, the second image, and a second hand skeleton based on the second image, wherein the one or more processes of the sign language recognition system comprise: a hand tracking process, a hand gesture detection process, and a sign language translation process.

[0109] Example 5 includes the method of example 1, further comprising: identifying a suggested word based on a combination of contextual information of the first augmented reality device, historical inputs of the user of the first augmented reality device, text corpus, and pre-defined terms, wherein the text indicates the suggested word.

[0110] Example 6 includes the method of example 5, further comprising: displaying the suggested word in the first display of the first augmented reality device; receiving a confirmation of the suggested word from the user of the first augmented reality device; and in response to receiving the confirmation, providing the suggested word to the second augmented reality device.

[0111] Example 7 includes the method of example 6, wherein receiving the confirmation comprising: detecting a predefined gesture from the user, the predefined gesture indicating the confirmation.

[0112] Example 8 includes the method of example 1, wherein distributing the one or more processes of the sign language recognition system further comprises: temporally or spatially distributing the one or more processes between the first augmented reality device and the second augmented reality device, and wherein a temporal distribution comprises processing alternating frames between the first augmented reality device and the second augmented reality device, wherein a spatial distribution comprises processing a first gesture from a first hand of the user only with the first augmented reality device and a second gesture from a second hand of the user only with the second augmented reality device.

[0113] Example 9 includes the method of example 1, wherein distributing the one or more processes of the sign language recognition system further comprises: detecting, by the first augmented reality device, an occlusion of the hand gesture in the first image; and in response to detecting the occlusion, assigning the second augmented reality device to perform the one or more processes based on the second image.

[0114] Example 10 includes the method of example 1, wherein the first augmented reality device includes a first head-wearable device, wherein the second augmented reality device includes a mobile hand-held device or a second head-wearable device.

[0115] Example 11 is a computing apparatus comprising: a processor; and a memory storing instructions that, when executed by the processor, configure the apparatus to: access a first image generated by a first camera of a first augmented reality device and a second image generated by a second camera of a second augmented reality device, the first image and the second image depicting a hand gesture of a user of the first augmented reality device; synchronize the first augmented reality device with the second augmented reality device; in response to synchronizing, distribute one or more processes of a sign language recognition system between the first augmented reality device and the second augmented reality device, wherein the one or more processes are

performed on a corresponding augmented reality device; collect results from the one or more processes from the first augmented reality device and from the second augmented reality device; and display, in near real-time in a first display of the first augmented reality device or in a second display of the second augmented reality device, text indicating a sign language translation of the hand gesture based on the results.

[0116] Example 12 includes the computing apparatus of example 11, wherein synchronizing the first augmented reality device with the second augmented reality device further comprises: map a first timestamp of the first image with a second timestamp of the second image.

[0117] Example 13 includes the computing apparatus of example 11, wherein synchronizing the first augmented reality device with the second augmented reality device further comprises: register a six-degree-of-freedom coordinate system of the first augmented reality device with a six-degree-of-freedom coordinate system of the second augmented reality device.

[0118] Example 14 includes the computing apparatus of example 11, wherein the instructions further configure the apparatus to: communicate the first image to the second augmented reality device, the first image providing an egocentric view of the hand gesture of the user of the first augmented reality device, and wherein the second augmented reality device is configured to perform the one or more processes of the sign language recognition system on a combination of the first image, a first hand skeleton based on the first image, the second image, and a second hand skeleton based on the second image, wherein the one or more processes of the sign language recognition system comprise: a hand track process, a hand gesture detection process, and a sign language translation process.

[0119] Example 15 includes the computing apparatus of example 11, wherein the instructions further configure the apparatus to: identify a suggested word based on a combination of contextual information of the first augmented reality device, historical inputs of the user of the first augmented reality device, text corpus, and pre-defined terms, wherein the text indicates the suggested word.

[0120] Example 16 includes the computing apparatus of example 15, wherein the instructions further configure the apparatus to: display the suggested word in the first display of the first augmented reality device; receive a confirmation of the suggested word from the user of the first augmented reality device; and in response to receiving the confirmation, provide the suggested word to the second augmented reality device.

[0121] Example 17 includes the computing apparatus of example 16, wherein receiving the confirmation comprising: detect a predefined gesture from the user, the predefined gesture indicating the confirmation.

[0122] Example 18 includes the computing apparatus of example 11, wherein distributing the one or more processes of the sign language recognition system further comprises: temporally or spatially distribute the one or more processes between the first augmented reality device and the second augmented reality device, and wherein a temporal distribution comprises process alternating frames between the first augmented reality device and the second augmented reality device, wherein a spatial distribution comprises process a first gesture from a first hand of the user only with the first augmented reality device and a second gesture from a second hand of the user only with the second augmented reality device.

[0123] Example 19 includes the computing apparatus of example 11, wherein distributing the one or more processes of the sign language recognition system further comprises: detect, by the first augmented reality device, an occlusion of the hand gesture in the first image; and in response to detecting the occlusion, assign the second augmented reality device to perform the one or more processes based on the second image.

[0124] Example 20 is a non-transitory computer-readable storage medium, the computer-readable storage medium including instructions that when executed by a computer, cause the computer to: access a first image generated by a first camera of a first augmented reality device and a second image generated by a second camera of a second augmented reality device, the first image and the

second image depicting a hand gesture of a user of the first augmented reality device; synchronize the first augmented reality device with the second augmented reality device; in response to synchronizing, distribute one or more processes of a sign language recognition system between the first augmented reality device and the second augmented reality device, wherein the one or more processes are performed on a corresponding augmented reality device; collect results from the one or more processes from the first augmented reality device and from the second augmented reality device; and display, in near real-time in a first display of the first augmented reality device or in a second display of the second augmented reality device, text indicating a sign language translation of the hand gesture based on the results.

[0125] Described implementations of the subject matter can include one or more features, alone or in combination as illustrated below by way of example.

Claims

1. A method comprising: accessing a first image generated by a first device and a second image generated by a second device, the first image and the second image depicting a hand gesture of a user of the first device; generating synchronized information that comprises first image metadata of the first image, first pose data of the first device, second image metadata of the second image, and second pose data of the second device; distributing one or more processes between the first device and the second device by sharing the synchronized information between the first device and the second device; accessing results from the one or more processes from the first device and from the second device; and displaying, in a first display of the first device or in a second display of the second device, information based on the results.
2. The method of claim 1, wherein generating the synchronized information comprises: mapping the first image metadata with the second image metadata.
3. The method of claim 1, wherein generating the synchronized information comprises: mapping a first timestamp of the first image with a second timestamp of the second image.
4. The method of claim 1, wherein generating the synchronized information comprises: registering a six-degree-of-freedom coordinate system of the first device with a six-degree-of-freedom coordinate system of the second device.
5. The method of claim 1, further comprising: communicating the first image to the second device, the first image providing an egocentric view of the hand gesture of the user of the first device, and wherein the second device is configured to operate a sign language recognition system on a combination of the first image and the second image, wherein the sign language recognition system comprises: a hand tracking process, a hand gesture detection process, and a sign language translation process, wherein the information comprises text indicating a sign language translation of the hand gesture based on an output of the sign language recognition system.
6. The method of claim 1, further comprising: identifying a suggested word based on a combination of contextual information of the first device, historical inputs of the user of the first device, text corpus, and pre-defined terms, wherein the information indicates the suggested word; displaying the suggested word in the first display of the first device; detecting a confirmation of the suggested word from the user of the first device; and in response to detecting the confirmation, providing the suggested word to the second device.
7. The method of claim 6, wherein detecting the confirmation comprises: detecting a predefined gesture from the user, the predefined gesture indicating the confirmation.
8. The method of claim 1, wherein distributing the one or more processes comprises: temporally or spatially distributing one or more processes between the first device and the second device, and wherein a temporal distribution comprises processing alternating frames between the first device and the second device, wherein a spatial distribution comprises processing a first gesture from a first hand of the user with the first device and processing a second gesture from a second hand of

the user with the second device.

9. The method of claim 1, wherein distributing the one or more processes comprises: detecting, by the first device, an occlusion of the hand gesture in the first image; and in response to detecting the occlusion, assigning the second device to perform the one or more processes based on the second image.

10. The method of claim 1, wherein the first device includes a first head-wearable device, wherein the second device includes a mobile hand-held device or a second head-wearable device.

11. A first device comprising: a processor; and a memory storing instructions that, when executed by the processor, configure the first device to perform operations comprising: accessing a first image generated by the first device and a second image generated by a second device, the first image and the second image depicting a hand gesture of a user of the first device; generating synchronized information that comprises first image metadata of the first image, first pose data of the first device, second image metadata of the second image, and second pose data of the second device; distributing one or more processes between the first device and the second device by sharing the synchronized information between the first device and the second device; accessing results from the one or more processes from the first device and from the second device; and displaying, in a first display of the first device or in a second display of the second device, information based on the results.

12. The first device of claim 11, wherein generating the synchronized information comprises: mapping the first image metadata with the second image metadata.

13. The first device of claim 11, wherein generating the synchronized information comprises: mapping a first timestamp of the first image with a second timestamp of the second image.

14. The first device of claim 11, wherein generating the synchronized information comprises: registering a six-degree-of-freedom coordinate system of the first device with a six-degree-of-freedom coordinate system of the second device.

15. The first device of claim 11, wherein the operations further comprise: communicating the first image to the second device, the first image providing an egocentric view of the hand gesture of the user of the first device, and wherein the second device is configured to operate a sign language recognition system on a combination of the first image and the second image, wherein the sign language recognition system comprises: a hand tracking process, a hand gesture detection process, and a sign language translation process, wherein the information comprises text indicating a sign language translation of the hand gesture based on an output of the sign language recognition system.

16. The first device of claim 11, wherein the operations further comprise: identifying a suggested word based on a combination of contextual information of the first device, historical inputs of the user of the first device, text corpus, and pre-defined terms, wherein the information indicates the suggested word; displaying the suggested word in the first display of the first device; detecting a confirmation of the suggested word from the user of the first device; and in response to detecting the confirmation, providing the suggested word to the second device.

17. The first device of claim 16, wherein detecting the confirmation comprises: detecting a predefined gesture from the user, the predefined gesture indicating the confirmation.

18. The first device of claim 11, wherein distributing the one or more processes comprises: temporally or spatially distributing one or more processes between the first device and the second device, and wherein a temporal distribution comprises processing alternating frames between the first device and the second device, wherein a spatial distribution comprises processing a first gesture from a first hand of the user with the first device and processing a second gesture from a second hand of the user with the second device.

19. The first device of claim 11, wherein distributing the one or more processes comprises: detecting, by the first device, an occlusion of the hand gesture in the first image; and in response to detecting the occlusion, assigning the second device to perform the one or more processes based on

the second image.

20. A non-transitory computer-readable storage medium, the computer-readable storage medium including instructions that when executed by a computer, cause the computer to: accessing a first image generated by a first device and a second image generated by a second device, the first image and the second image depicting a hand gesture of a user of the first device; generating synchronized information that comprises first image metadata of the first image, first pose data of the first device, second image metadata of the second image, and second pose data of the second device; distributing one or more processes between the first device and the second device by sharing the synchronized information between the first device and the second device; accessing results from the one or more processes from the first device and from the second device; and displaying, in a first display of the first device or in a second display of the second device, information based on the results.
